Do Your Friends Know Each Other?

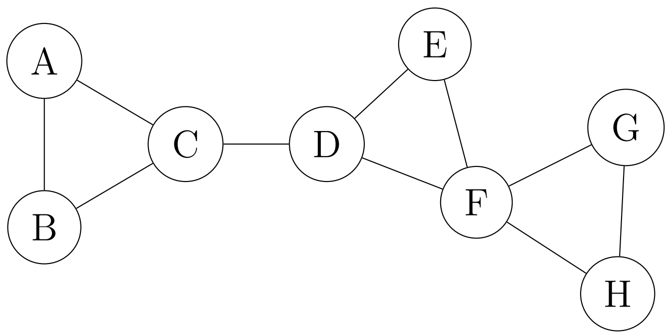
Question 1: Think of your two closest friends. Do those two know each other? ________

Now suppose we pick two people at random out of everyone at this university. Do *they* know each other? ________

Your two friends were not picked at random — they were picked by you. Write down, in one sentence, why you think that changes the answer.

Part 1: Counting the closed corners

Eight students share a dorm floor. Here is who eats lunch with whom.



Question 2: Take one student at a time. List their lunch partners, count how many *pairs* of those partners could possibly eat together, and then count how many of those pairs actually do. The row for A is filled in as an example.

Student	Partners	How many	Pairs possible	Pairs connected	Fraction
A	B, C	2	1	1	1
B					
C					
D					
E					
F					
G					
H					

Question 3: The last column is called the *local clustering coefficient* of a student. Based on the numbers you just wrote:

1. What does a value of 1 tell you about that student's social life?

   ____

2. What would a value of 0 tell you?

   ____

3. If a student has only one lunch partner, what should the value be — and does your formula give an answer at all?

   ____

Question 4: Average your eight fractions to get one number for the whole floor.

Average = ____________

Question 5: Now measure the same idea a completely different way, without looking at individual students.

1. Count the triangles on the floor — groups of three students who all eat with each other. ________
2. Count every "corner": a corner is a student together with two of their partners, whether or not those two eat together. (You already have this — it is the "pairs possible" column, summed.) ________
3. Every triangle gets counted as a closed corner three times, once at each of its three students. So the fraction of corners that are closed is

$$\frac{3 \times (\text{triangles})}{\text{corners}} = \underline{\hspace{3cm}}$$

Question 6: You now have two numbers that both claim to answer "do friends of friends know each other on this floor?", and they disagree.

1. Which one is larger? ________

2. Student F has four partners and a low fraction. How many times does F count in the average of Question 4? ________ How many corners does F contribute in Question 5? ________

3. So which of the two measures listens more to the popular students, and which treats everyone equally?

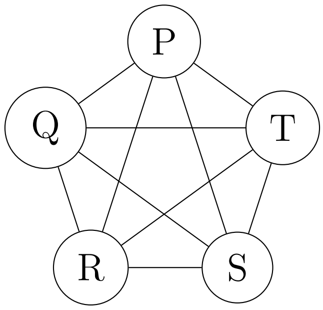
   ____

4. Two floors have identical numbers of triangles, but on one floor the triangles all sit around one very popular student. Which measure would notice the difference?

   ____

Part 2: Compared to what?

Question 7: Here are five students who *all* eat with each other.



1. Clustering of this group, by either recipe: ________
2. Average number of handshakes between two of them: ________
3. These are the best possible values: clustering as high as it goes, distance as short as it goes. Is this group a small world? Is it anything like a society?

   ____

4. So a high clustering number and a low distance number, on their own, prove ____________________.

Question 8: To judge the dorm floor we need something to judge it *against*. So build a floor with no social structure at all: take the same 8 students, consider all 28 possible lunch pairs, and toss a biased coin for each one, keeping the pair with probability p. Choose p so that this coin-flip floor has, on average, the same number of lunch pairs as the real one.

1. How many lunch pairs does the real floor have? ________ So $p =$ ________
2. On the coin-flip floor, take a student with 4 partners. Those partners form 6 possible pairs. Each of those 6 pairs is present with probability ________, so on average ________ of them are present, and that student's fraction from Question 2 comes out to ________.

3. Redo the previous item for a student with 2 partners, and one with 6.

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4. What is the clustering of a coin-flip floor, and what does it depend on? This is the punchline of Part 2 — say it in one sentence.

Question 9: Fill in the comparison. You already know how to compute distances by hand from the first worksheet, so the two distance numbers are given to you here.

	Real dorm floor	Coin-flip floor
Clustering (average of Q2)		
Clustering (corner recipe, Q5)		
Average handshakes	2.14	2.27

1. How many times more clustered is the real floor than the coin-flip floor? ________
2. How many times longer are its handshake chains? ________
3. Combine your two answers into a single number that is large when a network is far more clustered than chance without paying for it in longer chains. Write your recipe.

   ____

4. What value does your number take for the coin-flip floor itself? ________ What would it take for Ringville from the first worksheet — bigger or smaller than that? ________

Question 10: You used the average from Question 4. The student next to you used the corner recipe from Question 5. Compute the number you invented in Question 9 both ways.

Yours: ____________ Theirs: ____________

You will both publish a paper claiming the dorm floor is a small world. What must you report alongside the number for the claim to mean anything?

____

Question 11: Someone hands you a network you have never seen — a power grid, a brain, a citation network — and asks: is it a small world?

Write down the procedure you would run. You do not have to compute anything; just list the steps in order.